Scene-Specific Multiprototype Network for Remote Sensing Scene Graph Generation

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***Abstract*—Remote sensing scene graph generation (SGG) aims to capture both objects and their semantic relationships, offering a comprehensive understanding of complex scenes. However, two major challenges hinder the performance of existing methods. First, remote sensing images often contain a large number of objects, many of which are unrelated. Performing global feature interactions across all objects introduces noise from irrelevant pairs, degrading feature quality. Second, relationship categories in remote sensing scenes exhibit significant intraclass variation across different contexts, and a long-tailed distribution further complicates learning due to limited samples for tail classes. To address these issues, we propose the scene-specific multiprototype network (SSMPNet). Our method performs contextual interactions selectively based on object and relationship categories, reducing interference from irrelevant features. Moreover, we introduce a scene-specific multiprototype classification framework that better captures the diverse visual manifestations of each relationship class, while also improving discrimination under long-tailed distributions. The experimental results demonstrate that the proposed model achieves state-of-the-art (SOTA) performance, with a minimum improvement of 4.5% and a maximum improvement of 21.3% in mR@20 on the predicate classification (PredCls) task over baseline models.**

***Index Terms*—Relation comprehension, remote sensing, scene graph generation (SGG).**

I. Introduction

With the rapid advancements in satellite and drone technologies, a vast amount of high-quality remote sensing imagery has become accessible, leading to increasingly demanding requirements for image understanding [1], [2] and data analysis [3] in geographical science. Previously, the interpretation of remote sensing images primarily focused on holistic understanding tasks, such as scene classification [4], [5], [6], [7], [8] and change detection [9], [10], [11], [12], [13], as well as the understanding of isolated objects within the images, such as object detection [14], [15], [16], [17], [18] and semantic segmentation [19], [20], [21], [22], [23]. Indeed, these tasks focus on understanding remote sensing images at the pixel level, such as identifying objects, classifying scenes, or detecting changes, but they do not provide a comprehensive description of the events occurring within the images. While pixel-level analysis is crucial for extracting detailed information, it often falls short of capturing the broader context or the dynamic processes that define the events in the scene.

Based on object detection, we have identified the presence of objects such as airplanes and airports in remote sensing images. However, to further understand what events the two objects, airplane and airport, are jointly participating in and subsequently describe what is happening in the entire image, scene graph generation (SGG) [24], [25], [26], [27] is needed as an intermediate medium. The scene graph task can represent the objects, their attributes, and their relationships in a structured form, such as generating triplets like $<$ *airplane, park-at, airport* $>$ or $<$ *warship, sail-on, water* $>$.

By converting images into structured and semantically rich representations, scene graphs enable a deeper and more comprehensive understanding of visual content, significantly improving the ability to interpret and reason about complex visual scenes. SGG plays a vital role in a wide range of advanced computer vision applications, such as captioning for images and videos [28], [29], [30], [31], cross-modal retrieval [32], [33], [34], and visual question answering [35], [36], [37]. In the domain of natural scenes, extensive research has been conducted on scene graphs, including the development of various datasets [26], [27], [38] and the proposal of numerous methods [24], [39], [40], [41], [42], [43]. However, due to the inherent differences between remote sensing scenes and natural scenes, directly applying scene graph methods designed for natural scenes to remote sensing contexts is not appropriate. Remote sensing images often exhibit unique characteristics, such as larger spatial scales, diverse object distributions, and complex contextual relationships, which require tailored approaches to effectively generate scene graphs and achieve meaningful interpretation.

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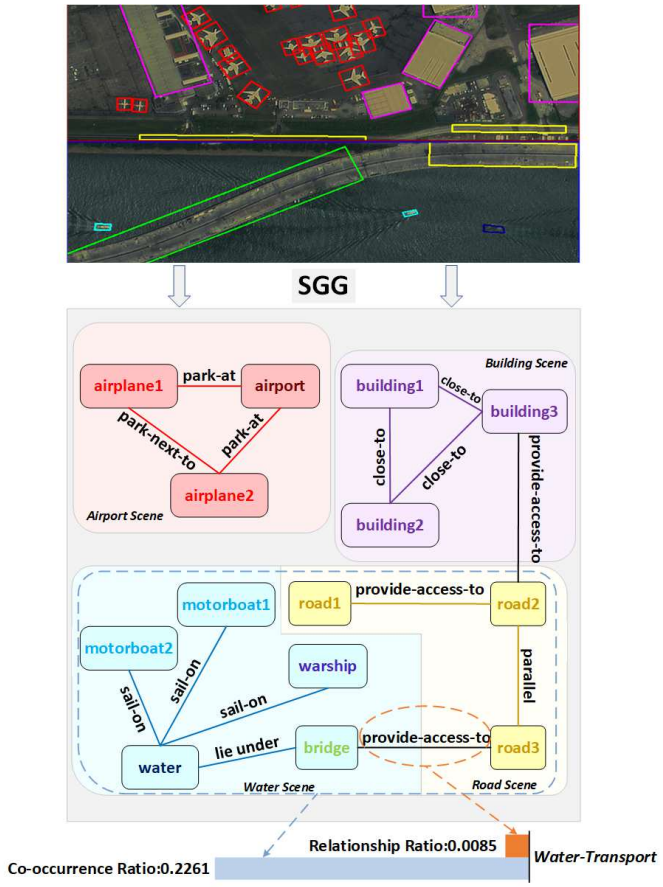


Fig. 1. Characteristic of multiple object classes co-occurring without relationships in a single remote sensing image. It presents an image with multiple scenes coexisting, such as airports, buildings, water activities, and roads, as well as the scene graphs formed by the relationships between objects. In the scene graph at the bottom, objects within the same color block belong to the same scene; for example, "airport" and "airplane" are categorized under the airport scene. It also presents a comparison of relationship ratios and co-occurrence ratios by category pairs.

Remote sensing images differ significantly from natural scene images, posing unique challenges for SGG methods. First, remote sensing images cover vast areas with distinctive scene attributes that can be divided into four categories: aviation scenes, water scenes, transportation scenes, and urban scenes. These images often contain multiple scenes within a single frame, such as airports and roads, or ports and residential areas. This coexistence of multiple scenes results in the relationships between objects being highly dependent on their respective scenes. For instance, Fig. 1 displays a coastal airport scene where objects from multiple scene categories, such as watercraft (e.g., motorboat), aviation facilities (e.g., airport), and transportation networks (e.g., road), coexist. However, relationships predominantly occur among objects within the same category, like <*motorboat, motorboat*> and <*motorboat, water*> interactions in the water-related scene, or < *airport, airplane* > associations in the aviation scene. Moreover, the probability of relationships between objects from different scene categories differs significantly. Relationships between urban and transportation objects (e.g., building–road) are much more likely to occur than those between aviation and water-related objects (e.g., airport–motorboat).

TABLE I

CO-OCCURRENCE OF OBJECTS WITH SCENE CLASSES 1 AND 2, AND THEIR CONTRIBUTION TO RELATIONSHIP PAIRS

entity scene class1	entity scene class2	$R_{\text{rel}}^{\text{s1-s2}}$	$R_{\text{co-occ}}^{\text{s1-s2}}$
transportation	urban	0.28	0.69
water	transportation	0.01	0.23
aviation	urban	0.01	0.22
water	urban	0	0.255
aviation	transportation	0	0.16
water	aviation	0	0.03

Additionally, Table I shows statistical analysis of the co-occurrence and relationships of objects from different scenes in the Recon1M dataset. $R_{\text{co-occ}}^{s1-s2}$ represents the proportion of images containing objects from both scene1 and scene2 (e.g., images contain ships belonging to water scene and airplanes belonging to aviation scene), and R_{rel}^{s1-s2} represents the proportion of relationships between objects from scene1 and scene2 (e.g., relationship triplets with the subject in the water scene and the object in the aviation scene)

$$R_{\text{co-occ}}^{s1-s2} = \frac{|\{I \in \text{Images} \mid \exists o_1 \in s1, \exists o_2 \in s2, o_1, o_2 \in I\}|}{|\text{Images}|}$$

$$R_{\text{rel}}^{s1-s2} = \frac{|\{(o_1, o_2) \in \text{Relation} \mid o_1 \in s1, o_2 \in s2\}|}{|\text{Relation}|}$$

where Images denotes the set of all images, Relation denotes the set of all object relationships, and o_1 and o_2 denote the objects. Statistical analysis of the Recon1M dataset reveals that over 22% of images contain objects from both water scenes and transportation scenes, but relationships formed between these two types of objects account for only 0.008%. This phenomenon indicates that cross-scene object relationships are sparse in remote sensing images, while intrascene object relationships are more prominent. However, in natural scenes, existing scene graph methods often overlook the scene attributes of objects, failing to differentiate between interactions based on scene attributes when modeling object–object or relationship–object interactions. This approach may induce interference in remote sensing scenes, compromising the accuracy of SGG. Second, when describing relationships between objects, there is often a certain bias. For example, when judging the relationship between "airplane" and "bus," we tend to focus more on the aviation scene in which the airplane is situated rather than the traffic scene to which the bus belongs. Therefore, we assign scene types to relationship pairs based on the scene types of the subject and object, and the distribution of relationship categories is closely related to the scene types (as shown in Fig. 2). For instance, the "moor-at" relationship accounts for only 3% of all relationships but makes up 29% in water scenes. This indicates that some relationship categories that are in the middle or tail of the overall distribution may be at the head of the distribution in specific scenes.

Additionally, the same relationship can exhibit significant differences across different scenes. Prototype-based matching classification [44] is a recently proposed effective method that captures the semantic features of relationships through prototype matching. However, during prototype optimization, to improve overall recognition accuracy and due to the imbalance in occurrence frequency across categories, head

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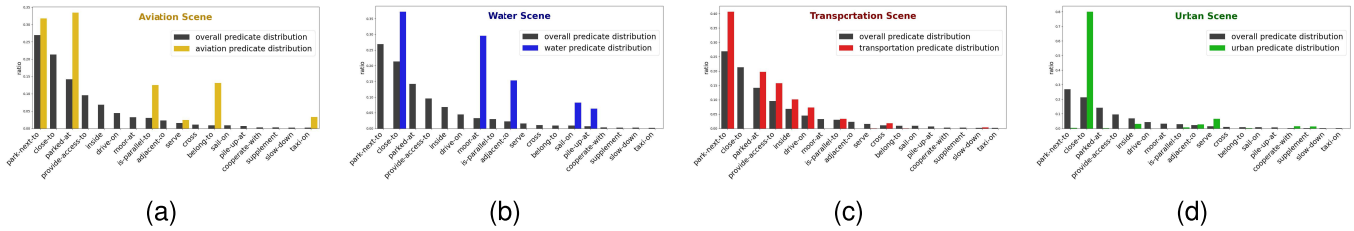


Fig. 2. Comparative distribution of relationship categories between the global dataset and four specific scenes. Relationships are categorized based on subject and object classes—for example, ⟨ship, harbor⟩ is classified as a water-related relationship pair. (a)–(d) Proportion of each relationship category within all relational triplets (brown bars) and within scene-specific subsets (yellow for water, red for aviation, blue for transportation, and green for urban). By comparing the global distribution with scene-specific distributions, we observe that several relationship categories that appear in the middle or tail of the global distribution become dominant within their corresponding scenes. This suggests that scene-specific categorization helps mitigate the issue where low-frequency (tail) relationship categories are misclassified as high-frequency (head) categories due to limited global occurrence.

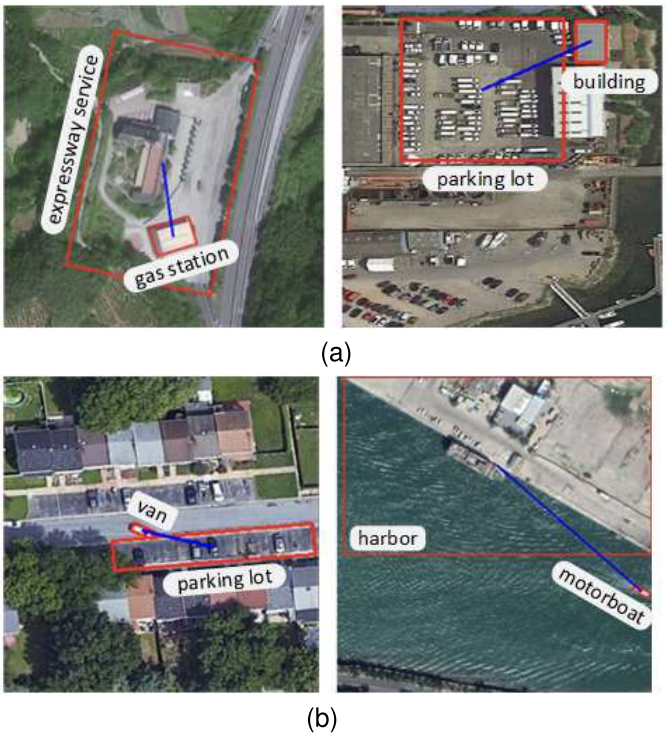


Fig. 3. Intraclass variations of relationship predicates across scenes. (a) "Serve" in the transportation scene and urban scene. (b) "Move away from" in the urban scene and water scene.

category prototypes often possess larger discriminative spaces, causing minority category features to be misclassified as head categories. Moreover, the same relationship category shows significant differences across various scenarios. As shown in Fig. 3, the "move away from" type often appears as cars leaving parking lots on land, while manifesting as ships departing docks on water. These two forms exhibit large differences in both spatial scale and visual appearance, making it difficult for a single prototype to represent such intraclass variations.

Therefore, specialized methods are needed to address the distinct challenges posed by remote sensing scenarios. We propose a scene-specific multiprototype network (SSMPNet), which includes two key components: the scene-aware contextual transformer (SACT) for feature encoding and the scene-aware multiprototype (SAMP) for adaptive prototype learning.

To overcome the first challenge, we propose a scene-specific context encoder that explicitly distinguishes between intra- and interscene interactions based on the scene attributes of objects and relationships. Specifically, this encoder separately models interactions within the same scene and interactions between different scenes, thereby better capturing the characteristics of object relationships in remote sensing images.

Meanwhile, to tackle the long-tail distribution problem and the issue of significant intraclass variations in relationships across different scenes, we propose a scene-specific multiprototype matching method. This method constructs different relationship prototypes based on different scene attributes to address the issues of imbalanced category distribution and large intraclass variations. Specifically, for each scene type, based on global semantic prototypes, we generate prototype offsets for relationship categories and determine the proportion of these offsets for different scenes based on scene attributes. This allows for the hierarchical construction of multiple scene-specific subprototypes, and various regularization is used to optimize the positioning of these subprototypes.

In summary, our main contributions are threefold.

1) We propose an SSMPNet that leverages the inherent scene attributes of objects and relationships to perform differentiated interactions and prototype construction. This approach is specifically designed to address the unique characteristics of remote sensing images, such as wide coverage and diverse scene compositions, achieving state-of-the-art (SOTA) performance on the remote sensing SGG task.
2) The SACT leverages scene-level context to model object relationships in remote sensing scenes, effectively reducing interference from irrelevant objects and relationships for more accurate SGG.
3) To tackle the challenges of large intraclass variations across different scenes and long-tail distribution of relationship categories, we propose SAMP that introduces scene-specific biases on top of global semantic prototypes.

The rest of this article is structured as follows. Section II critically reviews prior research and identifies unresolved challenges. Section III presents our methodological framework, with its architecture visualized in Fig. 4. Comprehensive experiments and comparative analyses are discussed in Section IV. Then, we discuss the impact of the proposed method and its current limitations in Section V. Finally,

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Section VI concludes with this work and proposes future research directions.

II. Related Works

In this section, we review some classic SGG methods, including message-passing approaches and debiasing techniques, as well as the latest methods for SGG in the remote sensing domain.

A. Message-Passing Methods in SGG

Message passing has long been fundamental to SGG, enabling information exchange between object nodes and relationship edges in graph-structured representations. It facilitates the fusion of local object features and global relational context, enhancing holistic scene understanding. Early graph-based models, such as Xu et al.'s [39] iterative message passing and Zellers et al.'s [40] Motif network, employed gated recurrent unit (GRU) and long short-term memory (LSTM) mechanisms, respectively, to propagate contextual information. Tang et al. [45] further introduced dynamic structures via TreeLSTM, while Yoon et al. [46] addressed semantic heterogeneity using relation-aware message passing on heterogeneous graphs. Other graph-based methods include [47], [48], [49], [50], and [51].

Moreover, transformer-based models have gained prominence due to their capacity to model long-range dependencies via self-attention. RTN [52] applied encoder–decoder architectures with novel edge encodings, while Jung et al. [53] proposed quad attention for fine-grained edge reasoning. Lin et al. [54] introduced edge-aware contextual refinement, and RelTR [55] simplified SGG into a set prediction task with coupled queries. These models excel in complex scenes by flexibly capturing diverse object–relation interactions beyond fixed graph structures.

B. Debiasing Methods

SGG often suffers from biased predictions due to long-tailed predicate distributions. To mitigate this, various debiasing strategies have been proposed.

Early approaches focused on rebalancing and context disentanglement. Tang et al. [56] introduced a causal framework using total direct effect to separate content and contextual bias. Dong et al. [57] proposed group collaborative learning, training multiple classifiers on predicate subsets. Yu et al. [58] used a hierarchical cognition tree to guide relationship prediction. Li et al. [59] employed a bipartite graph neural network (GNN) to enhance tail predicate representation, while CAModule [43] modeled causal relations via mediator variables to correct contextual bias.

More recently, prototype-based methods have gained traction. Zheng et al. [44] utilized semantic-aware prototypes to learn compact predicate representations. Jeon et al. [60] captured semantic diversity via region-based prototypes, and Chen et al. [61] proposed maintaining multiple prototypes per class to handle intraclass variation. These methods enhance generalization by modeling the semantic space of predicates, improving performance on underrepresented relations.

C. SGG in Remote Sensing

Remote sensing SGG is still in its early stages of research. Large-scale, high-quality scene graph datasets for remote sensing have only recently become available, such as STAR [62] and Recon1M [63]. Alongside these datasets, several methods tailored specifically for remote sensing scene graphs have emerged. Specifically, RPCM [62] adapts the message-passing mechanism from the IMP [39] method to model feature interactions between objects as well as between objects and relationships, followed by a prototype matching strategy to classify relation features. EGCAN [63] employs a more advanced transformer-based architecture to model object interactions. HMPNet [64] further improves upon the prototype matching module by introducing a hypergraph structure that fuses prototypes and captures associations between heterogeneous prototypes. However, as introduced in Section I, remote sensing images are characterized by the coexistence of multiple scenes and significant intraclass variation in relational features across different scenes. Although existing methods achieve promising results on remote sensing datasets, they do not specifically optimize for these unique characteristics of remote sensing data.

Specifically, for the first issue, we argue that feature interactions within scenes and across scenes should be treated differently—an aspect that has not been sufficiently considered in existing methods. RPCM models object and relationship feature interactions using a graph structure and employs several learnable matrices and to represent the importance of message passing among object–object, object–relationship, and relationship–relationship pairs. EGCAN [63] adopts a more powerful transformer module to encode all object features as a sequence using self-attention mechanisms. However, both approaches apply uniform message-passing mechanisms without distinguishing the importance of interactions between objects belonging to different scene categories. In contrast, our proposed SACT module explicitly differentiates interactions based on scene characteristics by first assigning scene attributes to entities and each relationship triplet according to the categories of the subject and object, and then employing multiple groups of transformers to perform differentiated feature interactions for triplets and entities belonging to different scene categories. This design enables more precise and context-aware relational reasoning, better capturing the scene-aware relationship distribution observed in remote sensing images.

For the second issue, we propose an SAMP matching strategy, motivated by the observation that the same relation type may exhibit different semantic and visual manifestations across scenes. Current prototype-based [44] classification methods have proven effective in alleviating class imbalance issues in datasets, and several approaches in remote sensing SGG, such as RPCM and HMPNet, have adopted this paradigm. Specifically, RPCM directly applies the method proposed in PENet, using relation label-derived text embeddings as base prototypes to match with relational features after the context messaging module, while HMPNet improves upon this by introducing a hypergraph structure to enhance connections

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between independent prototypes and further fusing image-modality and text-modality prototypes.

Despite their success, these methods fundamentally rely on single prototypes per relation category, which limits their ability to capture diverse semantic representations of the same relation across different scenes, often leading to suboptimal matching performance. In natural scene understanding, multiprototype strategies have also been explored; for instance, DPL [60] introduces semantic diversity by generating 40 additional prototypes through Gaussian noise added to the original base prototypes. However, during training, only the base prototype is actively optimized, while the subprototypes are derived from the base prototype and random noise, playing a role only during bias-free inference. Although this approach partially addresses semantic variation, the randomly generated subprototypes lack optimization and often require extensive tuning to achieve stable performance.

In contrast, the proposed SAMP module builds upon base semantic prototypes and generates relation category offsets and scene category offsets in a hierarchical manner, enabling the creation of scene-aware prototypes that better reflect the feature characteristics of different scenes. To ensure both semantic richness and prototype stability, we introduce multiple regularization techniques that maintain appropriate distances among: 1) base prototypes; 2) base and scene-specific prototypes; and 3) multiple prototypes within each scene category. Furthermore, we observe that certain relation categories that appear rare globally become dominant in specific scenes. By performing scenewise prototype matching, our method indirectly mitigates the bias toward high-frequency relations during model training, thereby improving generalization performance across both head and tail categories.

III. Methodology

In this section, we begin to introduce the scene-specific adaptive multiprototype network. The overall framework of the model is presented first. Following that, the section details the SACT module for relationship feature encoding and the SAMP module for matching and classification. The final section covers the model optimization strategy and loss function.

A. Overall Framework

The overall framework of SSMPNet is shown in Fig. 4. Our framework is divided into two main stages. In the first stage, a pretrained object detector is used to detect objects in the image, obtaining visual encoding features of the objects, oriented bounding boxes coordinates, object category prediction probabilities, and labels. In the second stage, relation pair features are extracted by obtaining image features from the union box of subject and object, and then concatenating these subject and object features. Subsequently, objects are classified into scene categories based on the object category prediction results. Then, the SACT module is used for scene-specific feature interaction between objects and relations, enabling contextual awareness and message passing to obtain encoded relation features. Finally, we construct an SAMP, matching relation features of different scene categories with corresponding scene prototypes to output predicted relation results.

B. Object Detection and Relation Feature Extraction

Our relationship detection framework begins with a comprehensive feature extraction pipeline that transforms raw visual inputs into rich object and relationship representations. Given an input image I, we employ a detection network to identify objects of interest, producing a set of N object proposals represented as $\{b_i, f_i^{\text{roi}}, l_i, P_i\}_{i=1}^{N}$, where $b_i \in \mathbb{R}^5$ denotes the bounding box parameters in the form of (x, y, w, h, θ), $f_i^{\text{roi}} \in \mathbb{R}^{4096}$ represents the extracted region of interest (ROI) features, $l_i \in \{1, 2, \ldots, C_{\text{obj}}\}$ is the predicted object class label, $P_i \in \mathbb{R}^{C_{\text{obj}}}$ contains the classification probabilities across all object categories, and $f_{ij}^{\text{union}} \in \mathbb{R}^{4096}$ represents the visual features extracted from the union region encompassing both objects in the relationship. The object feature enhancement module integrates multiple information sources through a series of neural network operations. For each detected object, we extract and process three key components:

1) visual features $f_i^{\text{roi}} \in \mathbb{R}^{4096}$ from the object ROI;
2) semantic embeddings $f_i^{\text{sem}} \in \mathbb{R}^{d_{\text{emb}}}$ derived from object classification;
3) geometric information $f_i^{\text{geo}} \in \mathbb{R}^{d_{\text{geo}}}$ encoded from bounding box coordinates.

For semantic embeddings, we utilize pretrained GloVe [65] word embeddings $E \in \mathbb{R}^{C_{\text{obj}} \times d_{\text{emb}}}$ with $d_{\text{emb}} = 300$ to incorporate linguistic priors

$$f_i^{\text{sem}} = \begin{cases} E_{l_i}, & \text{if using ground truth labels } l_i \\ \sum_{c=1}^{C_{\text{obj}}} P_i(c)\, E_c, & \text{if using softmax distribution.} \end{cases} \tag{1}$$

Simultaneously, we encode spatial information through a positional embedding network Φ_{pos} that transforms normalized bounding box coordinates into a geometric feature space. This network consists of a sequence of fully connected layers (FC_d) with batch normalization (BN) and rectified linear unit (ReLU) activation

$$\Phi_{\text{pos}} = \text{FC}_{128} \circ \text{ReLU} \circ \text{BN} \circ \text{FC}_{35} \circ \text{ReLU} \tag{2}$$

where $\circ$ denotes the sequential connection of modules, and

$$f_i^{\text{geo}} = \Phi_{\text{pos}}(\psi(b_i)) \in \mathbb{R}^{128}. \tag{3}$$

The function ψ extracts five geometric features from the bounding box parameters including normalized coordinates and angles.

These multimodal components are concatenated and processed through a hidden layer Φ_h, implemented as a multilayer perceptron (MLP)

$$\Phi_h = \text{FC}_{d_h} \circ \text{ReLU} \tag{4}$$

where $d_h = 512$ is the hidden dimension, and

$$f_i^{\text{obj}} = \Phi_h\left(\left[f_i^{\text{roi}}, f_i^{\text{sem}}, f_i^{\text{geo}}\right]\right) \in \mathbb{R}^{d_{\text{obj}}}. \tag{5}$$

This fusion creates augmented object features f_i^{obj} that capture visual attributes, semantic context, and spatial configurations.

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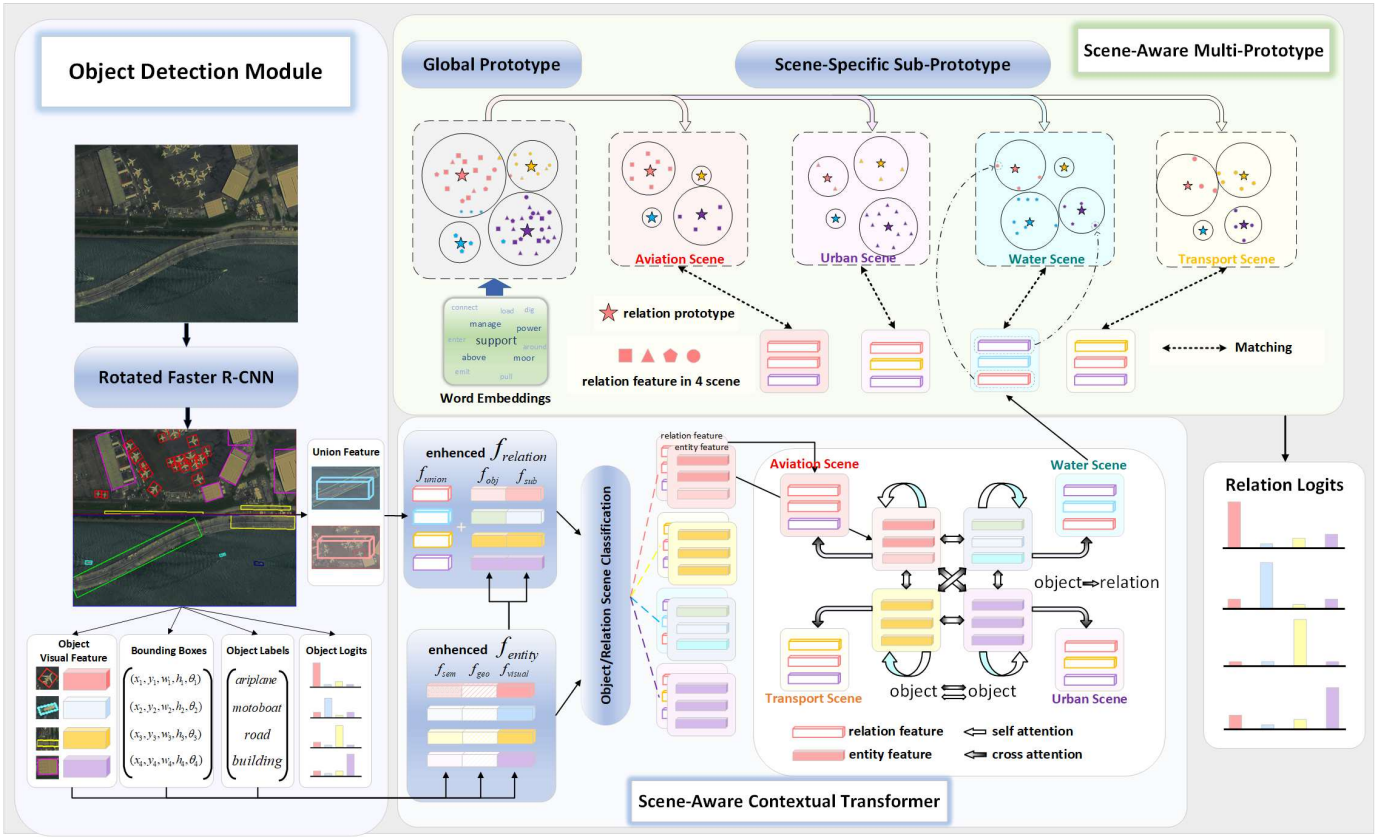


Fig. 4. Overall framework of the proposed SSMPNet is as follows. First, a detection module identifies objects in images and extracts their features. These features are, then, enhanced and divided into scene attributes based on the objects' categories. Next, the SACT module facilitates information interaction between object features of different scene attributes and between object and relationship features, producing encoded relationship features. Concurrently, SAMPs are hierarchically generated from global semantic prototypes. The encoded relationship features are matched with the corresponding scene prototypes to yield the final relationship classification.

For relationship representation, we adopt a dual-stream approach that combines object-pair information with union region features. Given the enhanced object features $\{f_i^{\text{obj}}\}_{i=1}^{N} \in \mathbb{R}^{d_{\text{obj}}}$ and union features $\{f_{ij}^{\text{union}}\}_{ij=1}^{M} \in \mathbb{R}^{d_{\text{roi}}}$ provided by the detector, we extract relationship features as follows.

First, we transform the object features to separate subject and object role representations

$$g_i^{\text{subj}}, g_i^{\text{obj}} = \Phi_{\text{split}}\left(f_i^{\text{obj}}\right) \tag{6}$$

where Φ_{split} consists of a dimension-adjusting MLP followed by a splitting operation, producing role-specific features $g_i^{\text{subj}}, g_i^{\text{obj}} \in \mathbb{R}^{d_{\text{obj}}}$. This crucial design enables the same object to have distinct representations depending on its role in a relationship, implicitly modeling the asymmetric nature of visual relationships.

For each relationship proposal (i, j), we construct a pairwise feature by concatenating the subject feature of object i with the object feature of object j

$$r_{ij}^{\text{pair}} = \left[g_i^{\text{subj}}; g_j^{\text{obj}}\right]. \tag{7}$$

We incorporate spatial configuration information by applying a spatial embedding function Φ_{spt} on the relative geometric features s_{ij} between the two objects

$$r_{ij}^{\text{pair}} = r_{ij}^{\text{pair}} \odot \Phi_{\text{spt}}\left(s_{ij}\right) \tag{8}$$

where s_{ij} encodes the relative position, size ratio, overlap, and other geometric relationships between objects i and j.

The final relationship representation is obtained by combining the processed object-pair features with the union region features through a fusion mechanism

$$f_{ij}^{\text{rel}} = \Phi_{\text{pair}}\left(r_{ij}^{\text{pair}}\right) + \Phi_{\text{union}}\left(f_{ij}^{\text{union}}\right) \tag{9}$$

where Φ_{pair} and Φ_{union} are the projection networks that align the features to a common dimension d_{rel}.

This dual-stream approach effectively integrates complementary information: the object-pair branch captures the semantic interaction between subject and object entities with implicit role awareness, while the union branch encodes the visual appearance of the relationship region. The resulting relationship features $f_{ij}^{\text{rel}} \in \mathbb{R}^{d_{\text{rel}}}$ and $f_i^{\text{obj}} \in \mathbb{R}^{d_{\text{obj}}}$ serve as the foundation for our subsequent scene-aware context transformer framework.

C. Scene Attribute Classification

After obtaining the enhanced object features and relation pair features from the preceding modules, we perform attribute classification on these features based on the subject–object categories. In Section III-C1–C1, we will introduce the proposed scene attributes for remote sensing objects and their corresponding relation pairs.

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1) Scene Category Definition: In remote sensing scene graphs, we categorize regions into four distinct scene types: aviation scenes, water scenes, transportation scenes, and urban scenes. This categorization provides a structured framework for classifying both individual objects and their pairwise relationships.

We divide remote sensing objects into these scene categories based on their typical occurrence patterns. Aviation scenes encompass objects such as *airplane*, *boarding bridge*, and *airport terminal*; water scenes include entities such as *passenger-ship*, *bridge*, and *harbor*. This object-scene mapping reflects the natural semantic groupings observed in remote sensing imagery.

2) Object-Level Scene Classification: The approach to object-level scene classification depends on the specific task setting. For the predicate classification (PredCls) task, where ground-truth object labels are provided, we directly map these fine-grained object labels to their corresponding scene categories using a predefined mapping function

$$c_i^{\text{obj_scene}} = \mathcal{M}\left(c_i^{\text{obj}}\right) \tag{10}$$

where c_i^{obj} is the ground-truth fine-grained category and $\mathcal{M}$ is the mapping from fine-grained categories to scene categories.

For the scene graph classification (SGCls) and scene graph detection (SGDet) tasks, where the model predicts object categories, we must determine scene categories based on predicted probability distributions over fine-grained classes. In these scenarios, we propose two complementary approaches.

a) Maximum probability mapping: This approach determines scene categories based on the most probable fine-grained class

$$c_i^{\text{obj}} = \text{argmax}_k P_i[k], c_i^{\text{obj_scene}} = \mathcal{M}\left(c_i^{\text{obj}}\right) \tag{11}$$

where P_i represents the probability distribution over all object categories produced by the pretrained detection model.

b) Mean probability aggregation: This approach accounts for uncertainty in fine-grained classification by aggregating probabilities across all classes within each scene category

$$p_i^{\text{scene}}[j] = \frac{1}{|\mathcal{C}_j|}\sum_{k\in\mathcal{C}_j} P_i[k], c_i^{\text{obj_scene}} = \text{argmax}_j p_i^{\text{scene}}[j] \tag{12}$$

where $\mathcal{C}_j$ represents the set of fine-grained categories associated with scene j. This approach is more robust to classification uncertainties and better captures the overall semantic context, particularly when fine-grained classifications have diffuse probability distributions.

3) Relationship Scene Classification: When classifying relationship pairs, we observe that humans exhibit inherent biases in their perception, often emphasizing one object's role over another in the relationship context. For instance, in relationship pairs like < *airplane, car* >, the *airplane* naturally assumes a more central position in the described event compared to the car. Similarly, when describing relationship pairs like < *bridge, road* >, the *bridge* serves as the primary guiding element. Furthermore, less frequently observed objects and their relationships tend to receive heightened attention from human observers. This cognitive bias informs our approach to relationship scene classification.

For relationships composed of objects from different scenes, we establish a scene sensitivity hierarchy based on object frequency and semantic prominence

$$S_{\text{Aviation}} > S_{\text{Water}} > S_{\text{Transportation}} > S_{\text{Urban}} \tag{13}$$

where S_{scene} represents the numerical sensitivity value of each scene, with higher values indicating higher priority of the scene category.

This hierarchy formalizes the observation that rarer and more specialized contexts (like aviation) typically dominate the semantic interpretation of mixed-scene relationships.

Formally, for each relationship pair (i, j) between objects with scene categories $c_i^{\text{obj_scene}}$ and $c_j^{\text{obj_scene}}$, we determine its scene category through a priority-based mapping function

$$c_{ij}^{\text{rel_scene}} = \Psi\left(c_i^{\text{obj_scene}}, c_j^{\text{obj_scene}}\right). \tag{14}$$

The mapping function Ψ implements the sensitivity hierarchy by assigning the relationship to the highest-priority scene category present among its constituent objects. Following our established hierarchy in (13), a relationship is classified according to the most semantically significant scene context of its participating objects.

Through this scene classification methodology, we organize both scene-specific object features $\{F_s^{\text{obj}}\}_{s=1}^4$ and relationship features $\{F_s^{\text{rel}}\}_{s=1}^4$ into four distinct scene categories. This scene-aware organization of visual elements provides the foundation for our subsequent scene-specific message passing and prototype learning, enabling the model to capture and leverage the unique characteristics of different remote sensing contexts.

D. Scene-Aware Contextual Transformer

After classifying objects and relationships into four scene categories, we leverage this hierarchical structure through a specialized transformer encoder (as shown in Fig. 5). This module processes both object and relationship features while preserving their scene-specific characteristics.

1) Relation Mask Prediction and Filtering: In SGG, not all proposed object pairs have meaningful relationships. To address this challenge, we employ a mask prediction mechanism that filters out spurious relationship proposals

$$M_{ij} = \Phi_{\text{mask}}\left(f_{ij}^{\text{rel}}\right) \tag{15}$$

where Φ_{mask} is a mask predictor that estimates the probability of a genuine relationship existing between objects i and j. The predictor combines local relationship features with global context

$$\Phi_{\text{mask}}\left(f_{ij}^{\text{rel}}\right) = \sigma\left(\Phi_2\left(\text{concat}\left[z_{ij}^{\text{local}}, z_{ij}^{\text{global}}\right]\right)\right) \tag{16}$$

where z_{ij}^{local} and z_{ij}^{global} are the local and global feature representations, Φ_2 is a multilayer prediction network, and σ is the sigmoid activation function.

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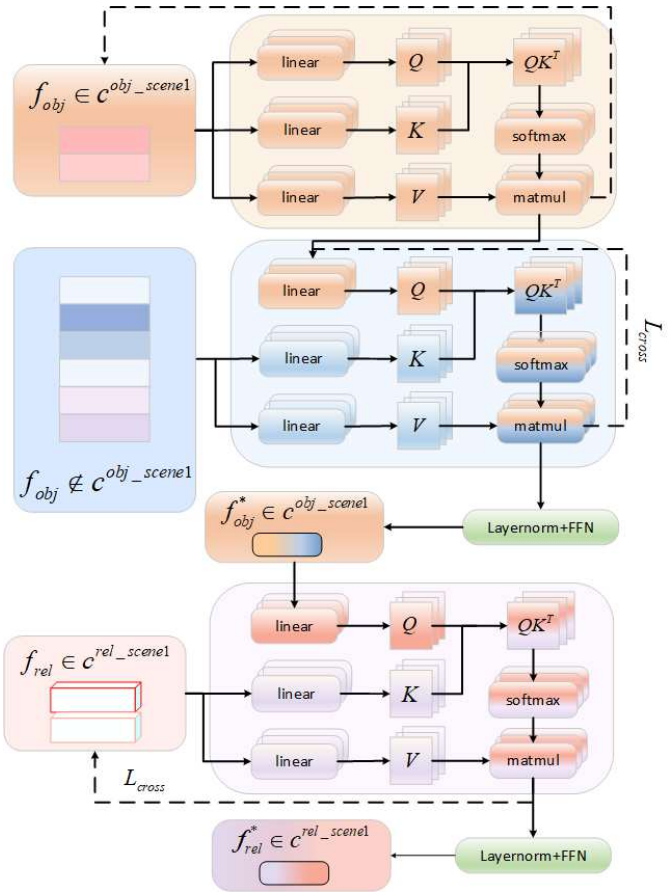


Fig. 5. SACT module is designed as follows. For object and relationship features belonging to scene category scene1, object interactions are handled in two steps: first, self-attention is applied within objects of scene category scene1; second, cross-attention is used between objects of scene category scene1 and those of other scene categories to aggregate global object information. Then, cross-attention is employed between objects and their corresponding relationship features within scene category scene1 to update the relationship features. This design minimizes noise from irrelevant objects by differentiating between intra- and interscene category interactions.

During training, we optimize this mask predictor using binary cross-entropy loss against ground-truth relationship existence labels

$$\mathcal{L}_{\text{mask}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \left[-Y_{ij}^b \log\left(M_{ij}^b\right) - \left(1 - Y_{ij}^b\right) \log\left(1 - M_{ij}^b\right) \right] \tag{17}$$

where $\mathcal{B}$ is a batch of relationship proposals, and Y_{ij}^b indicates whether a true relationship exists between objects i and j in the ground-truth annotations.

This approach prevents "contamination" of the feature space by spurious relationship proposals. By learning to distinguish between true and false relationships, the mask predictor helps focus computational resources on meaningful object interactions.

Based on these relationship existence scores, we select the top ρ fraction of relationships within each scene for feature updating

$$\mathcal{T}_s = \text{TopK}\left(\left\{M_{ij} | (i,j) \in F_s^{\text{rel}}\right\}, \lfloor \rho \cdot |F_s^{\text{rel}}| \rfloor\right) \tag{18}$$

where $\rho \in (0, 1]$ is a hyperparameter controlling the sparsity of the graph. This selective updating strategy ensures that only the most likely true relationships influence the feature refinement process.

2) Scene-Aware Message Passing: The core of our encoder is a scene-aware message passing mechanism that processes selected object and relationship features through multiple attention layers. For each scene s, we employ the following.

1) *Object Self-Attention:* Objects within the same scene exchange information through self-attention

$$\hat{f}_i^{\text{obj}_s} = f_i^{\text{obj}_s} + \left[\text{softmax}\left(\frac{Q^{\text{self}} \left(K^{\text{self}}\right)^T}{\sqrt{d_k}} \right) V^{\text{self}} \right]^{L_{\text{same}}}$$
$$f_i^{\text{obj}_s} = Q^{\text{self}} = K^{\text{self}} = V^{\text{self}} = \left\{ f_j^{\text{obj}} | c_j^{\text{obj_scene}} = s \right\} \tag{19}$$

where [] denotes the cascaded attention modules, and L_{same} represents the number of basic attention layers being stacked.

2) *Object Cross-Scene Attention:* Objects attend to objects from different scenes

$$\tilde{f}_i^{\text{obj}_s} = \hat{f}_i^{\text{obj}_s} + \left[\text{softmax}\left(\frac{Q^{\text{cross}} \left(K^{\text{cross}}\right)^T}{\sqrt{d_k}} \right) V^{\text{cross}} \right]^{L_{\text{cross}}}$$
$$Q^{\text{cross}} = \hat{f}_i^{\text{obj}_s}, K^{\text{cross}} = V^{\text{cross}} = \left\{ f_j^{\text{obj}} | c_j^{\text{obj_scene}} \neq s \right\}. \tag{20}$$

After the attention operations, we apply feed-forward networks (FFNs) with residual connections to integrate the learned features

$$f_i^{\text{obj}*} = \text{LayerNorm}\left(\tilde{f}_i^{\text{obj}} + \text{FFN}\left(\tilde{f}_i^{\text{obj}} \right) \right). \tag{21}$$

3) *Same-Scene Relationship-Object Attention:* Selected relationships attend to objects from the same scene

$$\tilde{f}_{ij}^{\text{rel}_s} = f_{ij}^{\text{rel}_s} + \left[\text{softmax}\left(\frac{Q^{\text{rel}} \left(K^{\text{rel}}\right)^T}{\sqrt{d_k}} \right) V^{\text{rel}} \right]^{L_{\text{self}}}$$
$$Q^{\text{rel}} = f_{ij}^{\text{rel}} = \left\{ f_k^{\text{rel}} | c_k^{\text{rel_scene}} = s \right\}$$
$$K^{\text{rel}} = V^{\text{rel}} = \left\{ \tilde{f}_k^{\text{obj}} | c_k^{\text{obj_scene}} = s \right\}. \tag{22}$$

3) Feature Integration: For relationships that were selected by the mask predictor, we update their features through

$$f_{ij}^{\text{rel}*} = \text{LayerNorm}\left(\tilde{f}_{ij}^{\text{rel}} + \text{FFN}\left(\tilde{f}_{ij}^{\text{rel}} \right) \right). \tag{23}$$

For relationships that were not selected (low mask scores), we preserve their original features to prevent noise propagation

$$f_{ij}^{\text{rel}*} = \begin{cases} \tilde{f}_{ij}^{\text{rel}}, & \text{if } (i,j) \in \mathcal{T}_{c_{ij}^{\text{scene}}} \\ f_{ij}^{\text{rel}}, & \text{otherwise.} \end{cases} \tag{24}$$

Through this selective, scene-aware encoding process, we enhance both object and relationship features in a hierarchical manner. Within each scene, objects and relationships first aggregate information through intrascene attention, capturing scene-specific contextual patterns (e.g., how aircraft interact with boarding bridges in aviation scenes). The subsequent cross-scene attention enables objects and relationships to leverage relevant information from other scenes (e.g., how aircraft-related relationships can benefit from understanding road–vehicle interactions in transportation scenes), thereby

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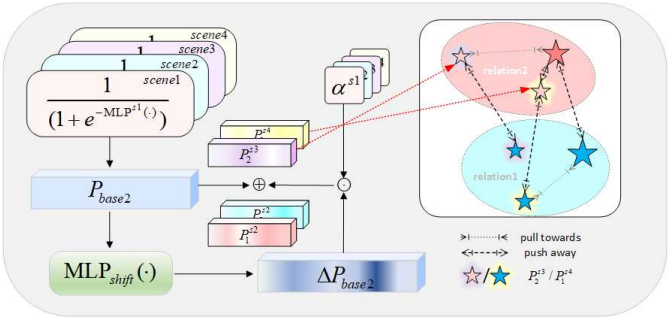


Fig. 6. Construction and optimization of multiscene prototype. Subscene prototypes are constructed through relation category offsets ΔP and per-scene adaptability scores α_s. In the optimization process, prototypes of different categories within the same scene push away from each other. Meanwhile, subscene prototypes of the same relation across different scenes maintain a proper level of similarity with the global prototype.

enriching the feature representations with complementary contextual information. This hierarchical attention mechanism ensures that our model can both preserve the unique characteristics of each scene and capture meaningful cross-scene dependencies, leading to more comprehensive relationship understanding in complex remote sensing scenarios. Through this process, we obtain the enhanced object features $f_i^{\mathrm{obj}*}$ and relationship features $f_{\mathrm{ij}}^{\mathrm{rel}*}$ that incorporate both scene-specific and cross-scene contextual information.

E. SAMP Learning

After obtaining the enhanced object and relationship features, we design an SAMP framework (Fig. 6) to capture the unique characteristics of relationships in different scenes. The key insight is that relationships in different scenes may share common semantic patterns while exhibiting scene-specific variations. For example, the relationship "next to" may have different visual manifestations when describing aircraft positions in aviation scenes versus vehicle arrangements in transportation scenes.

To address this dual nature of visual relationships, we propose a base prototype with adaptive scene-specific modifications

$$P_{\mathrm{base}} = \Phi_{\mathrm{pred}}(E_{\mathrm{rel}}) \tag{25}$$

where E_{rel} represents the relationship semantic embeddings that capture shared relationship concepts across scenes.

For each scene s, we introduce two key components to adapt the base prototype.

1) Relationship category shifts ΔP that capture the unique visual patterns in each relationship

$$\begin{aligned}\Delta P &= \mathrm{MLP}_{\mathrm{shift}}(P_{\mathrm{base}}) \\ &= W_2(\mathrm{ReLU}(W_1 P_{\mathrm{base}})).\end{aligned} \tag{26}$$

2) Adaptability scores α_s that control the degree of modification needed for each scene

$$\begin{aligned}\alpha_s &= \frac{1}{1+e^{-\mathrm{MLP}^s_{\mathrm{adapt}}(P_{\mathrm{base}})}} \\ &= \frac{1}{1+e^{-W_2^s(\mathrm{ReLU}(W_1^s P_{\mathrm{base}}))}}.\end{aligned} \tag{27}$$

The final scene-specific prototypes combine these components

$$P_s = P_{\mathrm{base}} + \alpha_s \odot \Delta P. \tag{28}$$

This design is motivated by several key observations.

1) Different scenes may require varying degrees of adaptation from the base semantic concepts. For instance, "connected to" might need minimal adaptation in transportation scenes (e.g., road connections) but significant modification in aviation scenes (e.g., aircraft–bridge connections).
2) Not all relationship categories require the same level of scene-specific modification. Some relationships may maintain similar patterns across scenes, while others might exhibit substantial scene-dependent variations.
3) The adaptability scores provide a learnable mechanism to automatically determine the appropriate balance between preserving common semantic patterns and incorporating scene-specific characteristics.

F. Multiscene Prototype Matching and Prototype Optimization

After constructing the scene-specific prototypes, we employ a prototype matching mechanism to measure the similarity between relationship features and their corresponding scene-specific prototypes. For a relationship instance with feature representation $f_{ij}^{\mathrm{rel}*}$ that belongs to scene $c_{ij}^{\mathrm{rel}-\mathrm{scene}}$, we compute its similarity with the scene-specific prototypes using the following formula:

$$S_{ij} = \tau \cdot \frac{f_{ij}^{\mathrm{rel}*} \cdot P^T_{c_{ij}^{\mathrm{rel}-\mathrm{scene}}}}{\|f_{ij}^{\mathrm{rel}*}\| \|P_{c_{ij}^{\mathrm{rel}-\mathrm{scene}}}\|} \tag{29}$$

where τ is a learnable temperature parameter that controls the sharpness of the similarity distribution. This cosine similarity-based approach ensures that the model focuses on the directional alignment between features and prototypes, rather than being influenced by their magnitudes. By evaluating each relationship feature against scene-specific prototypes corresponding to its coarse scene classification, we establish a context-aware matching framework that addresses the inherent visual variations across different scenes.

To optimize these prototypes during training, we design multiple objective functions that enhance their discriminative power while maintaining scene-specific adaptability.

1) Global Prototype Distinctiveness: To ensure that the base prototypes are sufficiently discriminative, we minimize the cosine similarity between different relationship prototypes

$$\mathcal{L}_{\mathrm{div}} = \frac{1}{K^2}\|\mathbf{S}_{\mathrm{base}}\|_{2,1} \tag{30}$$

where $\mathbf{S}_{\mathrm{base}} = \mathbf{P}_{\mathrm{base}}^{\top}\mathbf{P}_{\mathrm{base}}$ is the similarity matrix of base prototypes, and K is the number of relationship categories. This loss employs an $L_{2,1}$-norm regularization to reduce the overall similarity between prototypes, enhancing their discriminative capacity.

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2) Prototype Separation: Complementary to the cosine similarity-based loss, we also enforce a minimum Euclidean distance between prototype pairs

$$\mathcal{L}_{\text{dist}} = \frac{1}{K}\sum_{i=1}^{K}\max\left(0, \gamma_2 - \frac{1}{k}\sum_{j\in\mathcal{N}_i^k}\left\|\mathbf{p}_i - \mathbf{p}_j\right\|^2\right) \tag{31}$$

where $\mathcal{N}_i^k$ represents the set of k nearest prototypes to the ith prototype, and γ_2 is a margin parameter. This loss ensures that prototypes maintain sufficient separation in the embedding space, further enhancing their representational distinctiveness.

3) Scene-Specific Prototype Optimization: For each scene s, we employ three targeted optimization strategies.

1) *Intrascene Diversity:* Similar to the global diversity loss, but applied to each scene's prototype set separately

$$\mathcal{L}_{\text{div}}^s = \frac{1}{K^2}\|\mathbf{S}_s\|_{2,1} \tag{32}$$

where $\mathbf{S}_s = \mathbf{P}_s^\top \mathbf{P}_s$ is the similarity matrix of scene-specific prototypes. This loss ensures that relationships within each scene maintain distinct representations, but with a lower weight than the global diversity loss to allow for scene-specific adaptations.

2) *Scene-Global Alignment:* To prevent scene-specific prototypes from deviating too far from their semantic foundations, we enforce a soft constraint on their alignment with the base prototypes

$$\mathcal{L}_{\text{align}}^s = \text{MSE}\left(\frac{1}{K}\sum_{i=1}^{K}\frac{\mathbf{p}_i^s \cdot \mathbf{p}_i^{\text{base}}}{\|\mathbf{p}_i^s\|\|\mathbf{p}_i^{\text{base}}\|}, \alpha_{\text{target}}\right) \tag{33}$$

where α_{target} is a target similarity. This loss ensures that scene-specific adaptations preserve essential semantic properties while accommodating visual variations.

3) *Scene-Specific Contrastive Learning:* For each scene, we apply a contrastive loss that operates exclusively on samples from that scene

$$\mathcal{L}_{\text{cont}}^s = \frac{1}{N_s}\sum_{i=1}^{N_s}\max\left(0, d\left(f_i, p_{y_i}^s\right) - \frac{1}{k}\sum_{j=1}^{k} d\left(f_i, p_{n_j}^s\right) + \gamma_1\right) \tag{34}$$

where $d(\cdot,\cdot)$ is the squared Euclidean distance, N_s is the number of samples in scene s, $p_{y_i}^s$ is the prototype of the true class for sample i, $\{p_{n_j}^s\}_{j=1}^k$ are the k nearest negative prototypes, and γ_1 is a margin parameter. This loss ensures that features are closer to their corresponding class prototype than to other prototypes within the same scene, enhancing the discriminative power of scene-specific representations.

This adaptive prototype framework effectively balances the preservation of fundamental relationship semantics with the incorporation of scene-specific visual patterns, leading to more accurate and context-aware relationship recognition. By employing scene-specific prototype matching and carefully designed optimization objectives, our model can capture the nuanced variations of relationship appearances across different visual contexts while maintaining semantic consistency.

G. Joint Loss Functions

The overall training objective of our model incorporates multiple loss components that collectively optimize object detection, relationship prediction, and prototype representation. The joint loss function is formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{obj}} + \mathcal{L}_{\text{rel}} + \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{proto}} \tag{35}$$

where each component addresses a specific aspect of the SGG task.

1) Object and Relationship Classification Loss: These losses supervise object and relationship classification. Object classification uses refined object features from context modeling, while relationship classification uses similarity scores between relationship features and prototypes

$$\mathcal{L}_{\text{obj}} = \text{CE}\left(\hat{y}_{\text{obj}}, y_{\text{obj}}\right), \mathcal{L}_{\text{rel}} = \text{CE}\left(S, y_{\text{rel}}\right) \tag{36}$$

where CE is the cross-entropy loss, $\hat{y}_{\text{obj}}$ represents the object classification logits computed by combining initial classification logits with new logits derived from context-enhanced object features $f_i^{\text{obj}*}$, while S represents the prototype similarity scores, and y_{obj} and y_{rel} are the ground-truth object and relation labels, respectively. This refinement process creates a beneficial feedback loop, where relationship modeling enhances object recognition accuracy and the multiprototype representation is further refined through this similarity computation, encouraging the prototypes to be distributed more appropriately within the classification space for improved discriminability.

2) Mask Query Loss: As previously described, this loss ($\mathcal{L}_{\text{mask}}$) enhances the model's ability to focus on relevant regions for relationship recognition by supervising attention mask predictions with a binary cross-entropy objective. We weight this loss by a coefficient λ_{mask} to balance its contribution to the overall objective.

3) Prototype Optimization Loss: This loss combines various prototype optimization objectives

$$\mathcal{L}_{\text{proto}} = \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{dist}}\mathcal{L}_{\text{dist}} + \sum_{s=1}^{S}\left(\lambda_{\text{div}}^s\mathcal{L}_{\text{div}}^s + \lambda_{\text{align}}^s\mathcal{L}_{\text{align}}^s + \lambda_{\text{cont}}^s\mathcal{L}_{\text{cont}}^s\right) \tag{37}$$

where $\mathcal{L}_{\text{div}}$ and $\mathcal{L}_{\text{dist}}$ optimize global prototypes, while scene-specific losses ($\mathcal{L}_{\text{div}}^s$, $\mathcal{L}_{\text{align}}^s$, and $\mathcal{L}_{\text{cont}}^s$) ensure that each scene's prototypes are discriminative yet aligned with global semantics. We assign higher weights to global prototype losses compared to scene-specific losses to maintain robust base representations while allowing for scene-specific adaptations.

By jointly optimizing these complementary loss components, our model effectively learns to recognize objects, predict their relationships with scene awareness, and develop discriminative yet adaptable prototype representations. This integrated approach enables the generation of accurate and contextually appropriate scene graphs across various visual environments.

IV. Experiments

In this section, we present a comprehensive evaluation of our proposed scene-specific multiprototype framework. We

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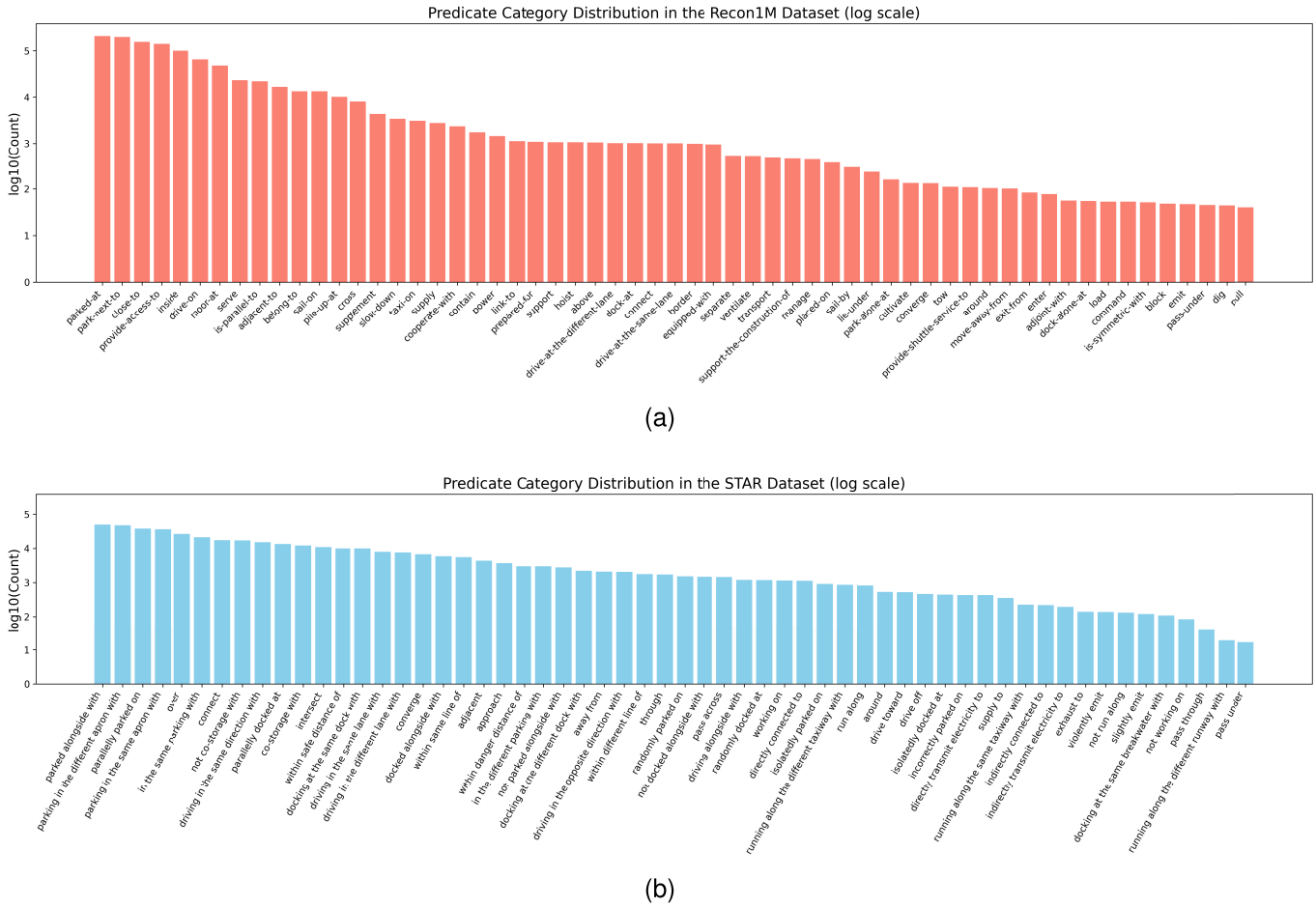


Fig. 7. Distribution of the number of instances per relation category in datasets: (a) Recon1M and (b) STAR.

first introduce the datasets, experimental setup, and evaluation metrics used in our experiments. Then, we compare our method with SOTA approaches and provide a detailed analysis of the results. Finally, we conduct extensive ablation studies to investigate the impact of different components and parameter settings, demonstrating the effectiveness of our proposed method for SGG tasks.

A. Datasets

1) Recon1M: Recon1M [63] is a comprehensive dataset specifically designed for remote sensing SGG tasks, comprising a total of 22 262 images. The object annotations span 60 categories across multiple scenes, including buildings, blocks, roads, intersections, vehicles, aircraft, control towers, ports, ships, and more. The relationship annotations cover 59 categories commonly encountered in remote sensing scenarios, including spatial relationships such as "parked on" and "parallel to," as well as high-level semantic relationships like "serving," "loading," and "towing."

This dataset presents particular challenges due to its complex object and relationship categories across diverse remote sensing scenes. Like natural scene datasets such as Visual Genome (VG) [26], Recon1M exhibits a significant long-tail distribution problem, as shown in Fig. 7(a), where a small number of categories dominate the dataset while many others have limited examples. For example, the relationship category "park at" appears 206 778 times in the dataset, while the more specific relationship "dock alone at" occurs only 54 times. In addition, the dataset includes certain relationships with broad semantic meanings, which exhibit substantial intraclass variation across different scenes, such as "serve" and "close to." These characteristics make Recon1M excellent for validating the effectiveness of our scene-specific multiprototype approach, which is specifically designed to handle diverse visual contexts and imbalanced category distributions.

2) STAR: STAR [62] is an SGG dataset designed for large-scale remote sensing images. The images in this dataset vary in size from 512 $\times$ 768 to 27 860 $\times$ 31 096 pixels, and the dataset includes 48 object categories and 58 relationship categories. Due to the large spatial coverage of each image, the number of objects per image can reach up to 6888, with diverse scenes such as airports, residential areas, and docks appearing within the same image. This poses significant challenges in modeling global object interactions across the entire image. As shown in Fig. 7(b), the dataset also presents substantial intraclass variation and a long-tailed distribution of relationship categories. For instance, the frequent relationship "parked alongside with" occurs 49 464 times, while the rare relation "pass under" appears only 16 times.

Therefore, both of the above datasets are specifically designed for the task of remote sensing SGG. They exhibit key characteristics of remote sensing images, including a

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large number of objects per image, complex object categories, and long-tailed distributions of relationship classes. These properties make the datasets well-suited for evaluating the effectiveness of our proposed method.

3) VG: The VG [26] dataset is one of the most commonly used benchmarks for SGG in natural image understanding. It contains a total of 108 077 images and defines 150 object categories and 50 relationship classes. We conduct experiments on this dataset to further validate the generalization capability of our method beyond the remote sensing domain.

B. Implementation Details

Recon1M Dataset: We follow previous settings and use a pretrained rotated Faster R-CNN [66] with ResNet-50 [67] backbone as the oriented object detector to obtain encoded features and predict object bounding boxes and categories. The parameters of the object detector are frozen during the SGG task. Our experiments are conducted on a single RTX 3090 GPU using the PyTorch [68] framework, with the Adam [69] optimizer, an initial learning rate of 0.0001, batch size of 4, and maximum training epochs of 24.

STAR Dataset: For fair comparison, we adopt HODNet, proposed in the original work, as the object detector. We strictly follow the original experimental settings: a batch size of 4, a learning rate of 0.00001, and a maximum of training 10 000 iterations.

For model parameter settings, we set the relationship selection ratio ρ to 0.7. We employ a dual-layer transformer attention mechanism for intrascene interactions between objects and between objects and relationships, while using a single layer for interscene interactions. The similarity target α_{target} between scene-specific prototypes and global prototypes is set to 0.7.

C. Tasks and Evaluation Metrics

1) Tasks: Following the standard protocol in SGG, we evaluate our method on three progressively more challenging tasks.

1) *PredCls:* Given ground-truth object bounding boxes and labels, the model predicts the relationships between object pairs. This task is considered the most fundamental and important in SGG research, as it directly evaluates the core relationship prediction capability without the confounding factors of object detection and classification. It serves as the primary benchmark for measuring a model's ability to understand visual relationships.
2) *SGCls:* Given ground-truth object bounding boxes, the model predicts both object labels and their relationships. This task evaluates the model's ability to recognize objects and their interactions simultaneously.
3) *SGDet:* The model detects objects and predicts their categories and relationships, starting from images alone. This is the most challenging and practical setting, reflecting real-world applications.

2) Evaluation Metrics: For evaluation metrics, we adopt the widely used Recall@K (R@K) and mean R@K (mR@K). Following the evaluation protocols of the respective datasets, we use $K = \{20, 50, 100, 500\}$ for Recon1M, and $K = \{1500, 2000\}$ for STAR. R@K measures the percentage of ground-truth relationships that appear in the top K most confident relationship predictions, which tends to favor frequent relationships due to the long-tail distribution of relationship categories. It is defined as the ratio of correctly identified positive predictions (true positive, TP) to the total number of actual positive instances (TP + false negative, FN)

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{38}$$

To better evaluate performance across all relationship categories, we also report mR@K, which computes the recall for each relationship category independently and then averages the results

$$\text{mR@K} = \frac{1}{C}\sum_{i=1}^{C} \text{R@K}_i \tag{39}$$

where C is the total number of relation categories, and R@K_i is the Recall@K for the ith category. This provides a more balanced assessment of the model's ability to recognize both common and rare relationship categories.

D. Main Results

We conducted experiments on the Recon1M and STAR, and compared our approach with previous SGG methods to validate its effectiveness.

Results on Recon1M Dataset:

1) Overall Performance: The results on the Recon1M dataset are presented in Table II. In the PredCls task, which focuses purely on relationship prediction with ground-truth object labels and bounding boxes, our SACT module achieves impressive recall scores of 62.3%/78.1%/87.1% on R@20/50/100, while SSMPNet excels in mean recall (mR) metrics with 51.6%/65.4%/74.0% on mR@20/50/100. These results indicate strong performance in both common and rare relationship prediction when object information is perfectly provided. For the more challenging SGCls task, where the model needs to predict object labels based on ground-truth bounding boxes, our approach maintains robust performance. SACT achieves 39.1%/48.3%/51.7% on R@20/50/100, and SSMPNet reaches 20.8%/26.8%/30.4% on mR@20/50/100. The performance decline from PredCls to SGCls is expected due to the additional challenge of object classification, yet our modules demonstrate strong resilience to this increased complexity.

In the SGDet task, which represents the most realistic scenario requiring both object detection and relationship classification, our approach shows remarkable robustness. SSMPNet achieves 16.0%/19.5%/22.7% on R@20/50/100 and obtains 10.3%/13.0%/14.6% on mR@20/50/100. Notably, the SSMPNet module maintains particularly strong mR performance in this challenging setting, demonstrating the effectiveness of scene-specific multiprototype learning when handling detection uncertainties. It is worth noting that while

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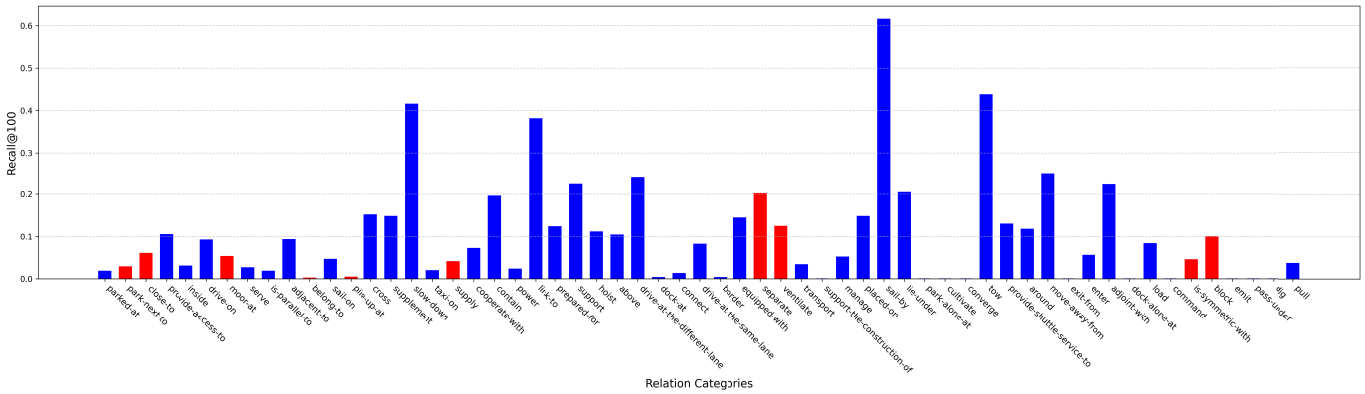


Fig. 8. Performance comparison between SSMPNet and PENet across relationship categories on PredCls tasks on Recon1M. The bars represent the differences in Recall@100 between SSMPNet and PENet, with blue bars indicating positive values (where SSMPNet outperforms PENet) and red bars showing negative values. SSMPNet exhibits notably enhanced discrimination capability for middle and tail categories, demonstrating its effectiveness in addressing the long-tail distribution problem in relationship recognition.

our method shows substantial improvements in the PredCls task, the performance gains in SGCls are relatively less pronounced. This is because under the moderate noise conditions of SGCls, all models maintain reasonably good performance, leaving less room for significant improvement. In SGCls, relationship prediction is inevitably affected by object classification errors, creating a cascading effect that limits the overall performance improvement regardless of how effective the relationship prediction module is. Additionally, incorrect object classification can lead to erroneous scene determination, as our object-scene classification module relies on accurate object recognition to properly categorize the scene context. This misclassification of scenes further impacts the effectiveness of both our scene-aware context modeling and scene-specific prototype matching. This observation aligns with findings in previous research that relationship recognition is heavily dependent on accurate object understanding. Nevertheless, our method still achieves consistent improvements across all tasks, demonstrating its robustness even under imperfect object recognition conditions. The truly remarkable performance of SSMPNet becomes evident in the high-noise SGDet task, where the ability to correctly classify tail relationship categories leads to substantial improvements in mR metrics.

2) Comparison With the SOTA Methods: We compare our proposed approach with previous SGG methods on the Recon1M dataset, with particular focus on the PredCls task, which directly evaluates relationship prediction capability.

a) Performance on recall metrics: In PredCls, the SACT module outperforms MOTIFS by 1.0%/1.0%/0.9% on R@20/50/100, which utilizes LSTM for sequential context modeling and struggles to capture the complex spatial relationships in remote sensing images. Similarly, SACT exceeds RTN by 1.1%/1.2%/1.2%, despite RTN's use of a transformer architecture for object context interaction. Compared to SQUAT, which selectively focuses on specific object–relation pairs for interaction, SACT shows improvements of 1.1%/1.9%/1.2%. Most importantly, SACT surpasses the recent SOTA EGCAN by 0.1%/0.1%/0.1%, which was specifically designed for remote sensing scenes with its geometric information embedding visual context-aware module, and still falls short of our scene-aware context modeling approach. Due to our scene-specific interaction strategy, object features and relationship features can more clearly aggregate information from within the same scene category through specialized attention mechanisms, while explicitly distinguishing between intrascene and cross-scene contextual patterns through separate interaction paths. This enables our model to adaptively weight and process information from different scene contexts, capturing both scene-specific relationship patterns and cross-scene interactions. Additionally, the SSMPNet also outperforms previous methods designed for natural scenes and achieves comparable results to the recent remote sensing-specific EGCAN, demonstrating the effectiveness and improvement of our approach for remote sensing scenarios.

b) Superior performance on mR metrics: Our methods achieve outstanding results on mR metrics, demonstrating powerful debiasing capabilities. Compared to the previous prototype-based debiasing method PE, our SACT module surpasses it by 1.3% and 2.8% on mR@20 and mR@100, respectively, using only improved relation feature encoding. The SAMP module shows even more significant improvements, increasing mR@20 and mR@100 by 6.0% and 7.5%, respectively, over PE, which was previously the model with the strongest debiasing capability. Fig. 8 illustrates the performance differences between the two methods across relationship categories. The results demonstrate that SSMPNet achieves substantial improvements in prediction accuracy for middle and tail relationships, with particularly notable gains in scene-dependent categories such as "sail-by" and "tow." By generating adaptable, scene-specific prototype shifts for each relationship category, and SSMPNet creates multiple specialized decision boundaries that better capture the visual variations within each relationship class. This is particularly beneficial for tail categories that may exhibit higher visual diversity relative to their limited training samples. Additionally, the adaptability mechanism dynamically adjusts how much each relationship category should adapt to scene-specific contexts, allowing for more flexible modeling of rare relationships.

Additionally, on the SGDet task, where all models experience a substantial performance decline due to the very high noise environment, SSMPNet still achieves significant

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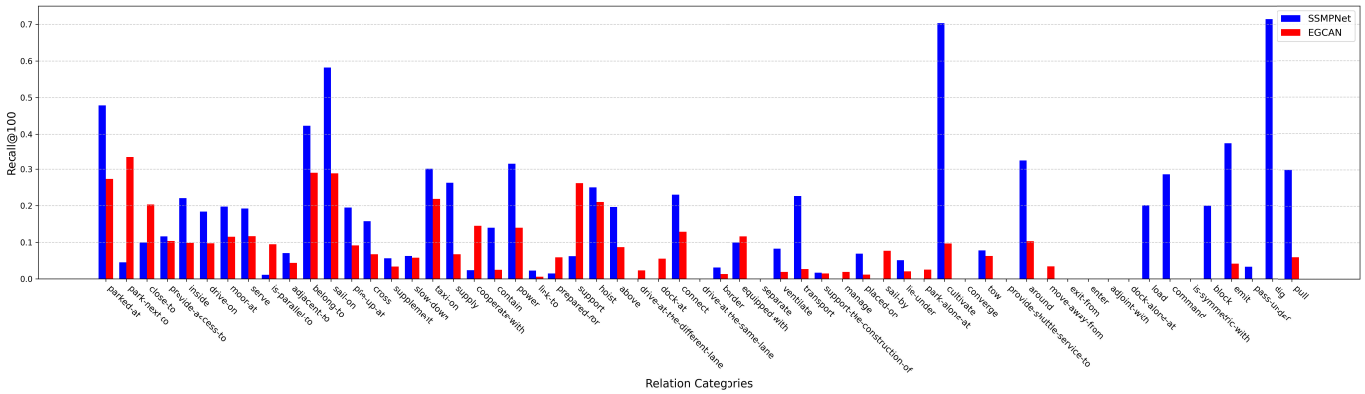


Fig. 9. Performance comparison between SSMPNet and EGCANet across relationship categories on SGDet task on Recon1M. The predictive performance of EGCAN is predominantly concentrated in head categories, exhibiting severe limitations in recognizing tail categories with near-zero accuracy. In contrast, SSMPNet demonstrates more robust performance, maintaining appreciable accuracy levels across the distribution, including tail categories.



(a) (b) (c) (d) (e)

Fig. 10. Visualization results of different models. (a) Oriented bounding boxes of objects. (b) Ground-truth scene graph. (c)–(e) Predicted results of PENet, EGCANet, and proposed SSMPNet, respectively. Red dashed lines, solid lines, and strike-through lines represent missing, incorrect, and redundant relationships.

improvements over EGCAN by 5.7%, 6.6%, and 6.8% on mR@20, mR@50, and mR@100, respectively, while outperforming bipartite graph neural network (BGNN) (the previous best model on this metric) by 1.8%, 2.5%, and 3.2%. As shown in Fig. 9, what distinguishes SSMPNet is its ability to correctly classify relationship categories in the tail of the distribution, resulting in more balanced prediction capabilities across all relationship classes. In the challenging SGDet task, relationship detectors must contend with inaccurate bounding boxes and object classes provided by the detector, causing

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TABLE II

OVERALL PERFORMANCE COMPARISONS WITH DIFFERENT METHODS ON PREDCLS, SGCLS, AND SGDET TASKS OVER RECON1M. SACT DENOTES USING RELATION FEATURES FROM SACT MODULE FOR DIRECT CLASSIFICATION, SSMPNET DENOTES CLASSIFYING ENCODED RELATION FEATURES BY SACT WITH MULTIPROTOTYPE MATCHING

Recall@K												
Method	PredCls				SGCls				SGDet			
	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500
IMP [39]	57.8	72.0	80.2	90.4	38.9	46.4	50.1	53.7	7.0	10.6	13.6	20.0
MOTIFS [40]	61.3	77.1	86.2	95.1	39.4	47.2	51.0	54.3	15.0	19.5	22.7	28.6
VtransE [70]	60.7	76.4	85.5	95.0	38.5	46.2	50.0	53.4	14.0	18.7	21.9	27.7
RTN [52]	61.2	76.9	85.9	95.4	39.2	47.1	51.0	54.5	15.0	19.7	23.0	28.7
GPSNet [54]	61.0	76.7	85.9	95.5	38.9	46.6	50.5	54.0	10.3	13.4	15.8	22.4
G-RCNN [71]	61.3	77.0	86.1	95.4	39.3	47.1	51.0	54.3	8.6	11.6	14.0	20.4
SQUAT [53]	61.2	76.2	84.5	94.7	37.2	44.6	48.5	52.8	13.5	17.3	19.8	24.7
BGNN [59]	61.0	76.3	85.3	95.2	35.8	42.9	47.0	51.5	15.1	16.9	17.7	18.3
PENet [44]	60.8	76.2	85.2	94.9	38.2	45.7	49.6	53.2	12.3	13.8	14.5	15.0
DPL [60]	60.1	75.4	84.1	94.1	38.0	42.4	47.6	52.2	13.3	15.8	17.3	19.8
EGCAN [63]	62.2	78.0	87.0	95.6	**40.2**	48.2	**52.0**	55.3	**16.5**	**21.5**	**24.7**	**30.0**
HMPNet [64]	61.1	76.8	86.2	95.2	38.8	46.3	50.8	54.7	13.5	14.9	18.7	21.7
SACT	**62.3**	**78.1**	**87.1**	**95.9**	39.1	**48.3**	51.7	**55.5**	16.3	20.2	23.4	28.4
SSMPNet	61.5	77.7	86.8	95.3	38.9	47.6	51.5	54.2	16.0	19.5	22.7	26.1

Mean Recall@K												
Method	PredCls				SGCls				SGDet			
	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500
IMP [39]	30.3	41.7	49.6	61.4	15.9	21.6	25.3	29.3	1.2	1.7	2.2	3.7
MOTIFS [40]	41.2	55.0	64.7	78.5	17.7	23.9	27.6	31.2	4.3	5.9	7.4	10.7
VtransE [70]	34.7	47.9	57.2	71.6	16.5	22.1	25.4	29.4	4.2	5.7	6.9	9.9
RTN [70]	42.0	55.6	64.4	77.9	19.1	24.5	27.9	32.0	3.5	5.2	6.6	10.0
GPSNet [52]	26.7	51.2	61.2	77.4	16.9	23.4	26.8	30.6	2.0	2.7	3.4	6.2
G-RCNN [71]	36.1	50.4	60.5	76.6	17.6	23.9	27.4	31.2	1.3	1.8	2.5	4.5
SQUAT [53]	38.5	51.4	60.5	75.6	15.0	20.0	23.7	28.0	3.3	4.3	5.2	7.3
BGNN [59]	37.8	51.0	50.6	76.4	13.6	18.8	22.0	25.8	8.5	10.5	11.4	12.8
PENet [44]	45.0	57.1	66.2	78.6	15.4	20.8	24.5	28.2	5.7	7.3	8.1	8.9
DPL [60]	44.8	56.4	65.7	78.4	15.0	20.1	24.0	27.4	5.2	7.1	7.8	8.2
EGCAN [63]	42.5	56.4	65.8	78.8	19.6	26.4	30.2	34.0	4.6	6.4	7.8	10.7
HMPNet [64]	47.1	58.3	67.7	79.0	19.2	22.1	25.7	29.6	7.1	8.3	10.2	11.4
SACT	46.3	60.0	69.0	81.9	17.6	23.3	26.8	30.3	8.0	9.9	10.8	12.1
SSMPNet	**51.6**	**65.4**	**74.0**	**84.3**	**20.8**	**26.8**	**30.4**	**34.2**	**10.3**	**13.0**	**14.6**	**16.9**

severe performance degradation across all methods. Even among correctly identified relationships, most models predominantly predict head categories, almost entirely misclassifying tail relationships. Consequently, while models can achieve acceptable recall metrics, they typically perform poorly on mR metrics that assess balanced recognition across all categories. SSMPNet addresses this limitation by creating specialized prototypes for different visual manifestations within each relationship category, enabling more accurate classification of tail relationships even under very highly noisy detection conditions, thereby achieving a more balanced performance across the entire relationship distribution.

3) Visualization Analysis: To provide an intuitive understanding of our method's effectiveness, we visualize SGG results from different approaches. Fig. 10 shows qualitative examples comparing our method with baseline approaches. Our model can correctly identify rare relationships with scene-specific characteristics, such as "tow" and "sail-by." It also reduces misjudgments between unrelated objects, especially when objects belonging to two different scene categories are in close proximity, such as the incorrect relationship < *road, belong-to, airport* >.

These visualizations confirm our quantitative findings and provide additional insights into how our dual-path approach addresses the challenges of remote sensing SGG.

a) Results on star dataset: To validate the generalization of our method in remote sensing scenarios, we conducted additional experiments on another widely used dataset STAR. The results are summarized in Table III. Consistent with its performance on Recon1M, our method also achieves competitive results on STAR. While maintaining a comparable overall recall rate, our approach significantly improves the mR, indicating enhanced recognition capability for tail categories with minimal degradation in performance on head categories.

Specifically, under the PredCls setting, our method outperforms the latest SOTA model RPCM by 1.3% and 1.6% in mR@1500 and mR@2000. Compared to the previous models with the highest overall performance, GPSNet and SHA, our method (SSMP) only incurs a minor drop of 0.1% and 0.3% in R@1500 and R@2000, while achieving substantial improvements of 9.5% and 9.1% in mR@1500 and mR@2000, respectively. These results further demonstrate the effectiveness of our approach in balancing performance across both frequent and rare relation categories.

b) Results on VG dataset: To further validate the effectiveness of the proposed SAMP approach, we evaluate the SAMP module on the natural image dataset VG. For the object categories in the dataset, we group them into two main classes: Animals (including humans, denoted as A) and Products (denoted as P). Based on the subject–object categories, we classify the relationship triplets into three types: AA, AP, and PP. Correspondingly, three types of subprototypes are generated from each basic semantic prototype to model these relationship patterns separately.

We employ Motifs as the relationship feature extractor to compare the performance of different prototype-based matching approaches. The experimental results are presented in Table IV.

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TABLE III
OVERALL PERFORMANCE COMPARISONS WITH DIFFERENT METHODS ON PREDCLS, SGCLS, AND SGDET TASKS OVER STAR DATASET

Method	PredCls				SGCls				SGDet			
	R@1500	R@2000	mR@1500	mR@2000	R@1500	R@2000	mR@1500	mR@2000	R@1500	R@2000	mR@1500	mR@2000
IMP [39]	51.5	53.2	21.5	22.3	46.1	47.5	18.6	19.4	18.1	19.0	5.6	6.0
MOTIFS [40]	62.0	64.0	29.4	30.6	50.4	52.3	24.2	25.4	20.6	21.6	7.8	8.2
GPSNet [54]	**65.4**	66.9	32.5	34.1	29.2	30.4	14.3	15.0	24.2	25.0	9.5	10.1
SHA [57]	65.0	**67.0**	33.3	34.8	51.0	52.6	23.3	24.2	28.0	29.1	10.2	10.8
HETSGG [46]	63.8	65.4	29.7	31.0	27.6	29.0	12.1	12.8	19.5	20.3	5.4	5.7
PENet [44]	62.9	65.0	37.0	38.3	41.8	43.5	20.6	21.6	21.3	22.5	8.8	9.3
HMPNet [64]	62.9	65.1	37.5	38.6	48.3	49.7	25.7	26.1	25.5	26.3	10.2	11.0
RPCM [62]	64.2	65.9	41.2	42.3	51.3	52.7	**30.0**	**30.9**	27.2	28.5	**11.5**	12.1
SSMPNet	65.3	66.7	**42.5**	**43.9**	**52.7**	**53.9**	29.3	30.4	**28.2**	**29.6**	11.2	**12.3**

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT PROTOTYPE MATCHING METHODS ON PREDCLS, SGCLS, AND SGDET TASKS OVER THE VG DATASET

Method	PredCls	SGCls	SGDet
	mR@50/100	mR@50/100	mR@50/100
MOTIFS+PR [44]	31.1/33.5	17.7/18.5	12.0/14.3
MOTIFS+HMP [64]	32.5/34.7	18.1/19.6	12.5/14.7
MOTIFS+DPL [60]	33.7/37.4	**18.5**/20.1	13.0/15.6
MOTIFS+SAMP	**34.6/37.9**	18.3/textbf20.5	**13.2/15.9**

As shown in the experimental results, our method also achieves competitive performance on the VG dataset compared to other SOTA approaches. Compared to the baseline single-prototype matching method, SAMP improves the mR@50 and mR@100 by 3.5% and 4.4%, respectively, under the PredCls setting. Moreover, compared to DPL, a method specifically designed for natural images, SAMP achieves improvements of 0.9% and 0.5% on mR@50 and mR@100, respectively, demonstrating the general effectiveness of our scene-specific multiprototype design in addressing the challenges posed by long-tailed data distributions. In addition, the experimental results indicate that multiprototype methods such as DPL and SAMP exhibit better debiasing effects than single-prototype approaches like HMPNet, especially in improving performance on rare relation categories without significantly sacrificing accuracy on frequent ones.

E. Ablation Study

To thoroughly analyze the effectiveness of each component in our proposed approach, we conduct a series of ablation experiments on the Recon1M dataset. These experiments provide insights into the contribution of different design choices and validate the effectiveness of our key innovations.

1) Effectiveness of SACT and SAMP Modules: First, we evaluate the effectiveness of the SACT and SAMP modules through targeted ablation experiments. We test the SACT + PE and SAMP model variants across all three tasks. Here, SACT + PE refers to applying single-prototype matching (PE) for classification on relationship features encoded by the SACT module, while SAMP refers to directly applying multiprototype matching classification to the original relationship features f_{ij}^{rel} without SACT encoding. These variants enable us to clearly demonstrate the individual contributions of the SACT module and the SAMP module to the overall performance.

As shown in Table V, our proposed modules effectively address three key challenges in visual relationship recognition. The SACT module enhances relationship features by increasing interclass distances in the feature space, as demonstrated by the improvement from PE to SACT + PE (from 60.8% to 61.5% in R@20 and 45.0% to 49.0% in mR@20 on PredCls). Meanwhile, SSMPNet tackles two additional challenges: it captures the scene-specific variations of visual relationships through multiprototype matching and adapts to the distinct category distributions in different scenes where long-tail relationships may become dominant in specific contexts. By combining these advantages, our full model achieves substantial improvements over the baseline PE by 0.7% R@20 and 6.6% mR@20 on PredCls. The significant gain particularly demonstrates our method's effectiveness in handling tail categories through enhanced feature representation and scene-specific modeling without compromising the performance on head categories.

2) Scene Classification Methods: We investigate the impact of different scene classification strategies on our modules' performance. We compare two approaches: maximum probability classification, which assigns the scene category based on the class with the highest probability, and average probability classification, which calculates the average probability for each scene category and selects the one with the highest average.

The results in Table VI demonstrate that average probability classification consistently outperforms maximum probability classification across most modules and tasks. This suggests that considering the collective evidence from all classes within a scene category provides a more robust scene assignment than relying solely on the individual class with the highest confidence. The improvement is particularly significant in the SGDet task, where object detection uncertainty could otherwise lead to incorrect scene assignments when using only the maximum probability class.

3) Impact of Scene Category Grouping: We explore different scene category grouping strategies to understand the optimal granularity for scene-aware modeling: 1) four categories: aviation, water, transportation, and urban; 2) three categories: aviation, water, transportation + urban; and 3) two categories: water, aviation + transportation + urban.

As shown in Table VII, we evaluate the impact of different scene category granularities on SSMPNet's performance. Our results demonstrate a clear trend: increasing the number of scene categories consistently improves performance across all metrics, with particular gains in mR. When increasing from

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TABLE V
ABLATION STUDY RESULTS OF SACT + PE, SAMP, AND SSMPNET

Method	Recall@K											
	PredCls				SGCls				SGDet			
	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500
SACT+PE	61.5	77.1	86.2	**95.7**	38.5	47.2	51.4	53.9	15.1	19.0	20.3	25.3
PENet	60.8	76.2	85.2	94.9	38.2	45.7	49.6	53.2	12.3	13.8	14.5	15.0
SAMP	61.2	77.5	86.5	95.6	38.6	47.6	51.0	**54.3**	15.7	19.2	21.5	25.7
SSMPNet	**61.5**	**77.7**	**86.8**	95.3	**38.9**	**47.6**	**51.5**	54.2	**16.0**	**19.5**	**22.7**	**26.1**

Method	Mean Recall@K											
	PredCls				SGCls				SGDet			
	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500
SACT+PE	49.0	61.3	69.4	81.3	18.7	24.7	29.7	33.5	8.9	11.5	12.6	14.7
PENet	45.0	57.1	66.2	78.6	15.4	20.8	24.5	28.2	5.7	7.3	8.1	8.9
SAMP	50.9	63.4	72.4	83.5	20.1	25.5	30.0	34.2	9.5	12.4	14.0	16.1
SSMPNet	**51.6**	**65.4**	**74.0**	**84.3**	**20.8**	**26.8**	**30.4**	**34.2**	**10.3**	**13.0**	**14.6**	**16.9**

TABLE VI
COMPARISON OF SCENE CLASSIFICATION METHODS ACROSS SACT AND SAMP MODULES

Methods	Model	Recall@K							
		SGCls				SGDet			
		R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500
SACT	max	38.9	47.5	51.2	54.1	15.7	**20.3**	22.8	27.4
	mean	**39.1**	**48.3**	**51.7**	**55.5**	**16.3**	20.2	**23.4**	**28.4**
SSMPNet	max	38.1	46.5	50.7	53.6	15.8	19.0	**22.8**	25.6
	mean	**38.9**	**47.6**	**51.5**	**54.2**	**16.0**	**19.5**	22.7	**26.1**

Methods	Model	Mean Recall@K							
		SGCls				SGDet			
		mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500
SACT	max	17.1	22.8	26.0	29.7	7.4	9.7	10.2	11.4
	mean	**17.6**	**23.3**	**26.8**	**30.3**	**8.0**	**9.9**	**10.8**	**12.1**
SSMPNet	max	20.4	26.3	29.2	32.3	9.7	12.5	14.2	16.3
	mean	**20.8**	**26.8**	**30.4**	**34.2**	**10.3**	**13.0**	**14.6**	**16.9**

TABLE VII
ABLATION STUDY ON THE NUMBER OF SCENE CATEGORIES FOR SAMP ON PREDCLS TASK

Scene	Recall@K				Mean Recall@K			
	20	50	100	500	20	50	100	500
2	61.1	76.7	86.0	94.9	47.4	61.2	70.0	83.2
3	61.4	77.2	86.2	95.0	50.4	62.4	70.3	83.7
4	**61.5**	**77.7**	**86.8**	**95.3**	**51.6**	**65.4**	**74.0**	**84.3**

TABLE VIII
ABLATION STUDY ON TARGET SIMILARITY BETWEEN SCENE-SPECIFIC AND GLOBAL PROTOTYPES

α_{target}	Recall@K				Mean Recall@K			
	20	50	100	500	20	50	100	500
0.3	61.2	76.8	85.8	95.0	47.4	61.9	70.9	83.0
0.5	61.3	76.8	85.8	95.2	49.2	62.3	71.2	83.6
0.7	**61.5**	**77.7**	**86.8**	**95.3**	**51.6**	**65.4**	**74.0**	**84.3**
0.9	61.0	76.7	85.7	95.1	49.2	62.9	71.2	83.4

two to three scene categories, we observe improvements of 0.3% in R@20 and 3.0% in mR@20. Further increasing to four categories yields additional gains of 0.1% in R@20 and 1.2% in mR@20. The total improvement from two to four categories reaches 0.4% in R@20 and 4.2% in mR@20, with the four-category configuration achieving the best performance.

From both the contextual interaction and multiprototype matching perspectives, progressively increasing scene categories enhances the model's capacity to capture scene-specific patterns, leading to consistent performance gains. In the SACT module, finer categorization refines contextual modeling: with only two categories, diverse environments (aviation, transportation, and urban) are forced to share a single attention mechanism, causing contextual confusion; separating aviation into its own category at the three-category stage enables specialized modeling of aircraft-related interactions; and further dividing into four categories allows each scene type—aircraft on runways, ships at docks, vehicles on roads, and buildings in urban areas—to preserve its unique contextual structure. Similarly, in the SAMP module, increased granularity reduces prototype competition and promotes specialized prototype learning: with two categories, relationships are crowded into a shared prototype space; isolating aviation at three categories improves mR@20 by 3.0%; and distinguishing transportation and urban scenes at four categories yields an additional 1.2% gain by enabling each to develop dedicated prototype representations. Together, these mechanisms demonstrate that finer scene division progressively disentangles contextual and relational patterns, resulting in more accurate and discriminative scene understanding.

4) Similarity Between Scene Prototypes and Main Prototype: In Section III-F, we propose the scene-global alignment loss $\mathcal{L}^s_{\text{align}}$ to constrain the divergence between scene-specific prototypes and global semantic prototypes. This loss ensures that scene-specific prototypes maintain appropriate connections with global semantic knowledge while preserving their ability to capture scene-specific features. We investigate the impact of different similarity constraints by varying the target similarity parameter α_{target} between scene-specific and global prototypes.

As shown in Table VIII, the results demonstrate that a moderate similarity constraint ($\alpha_{\text{target}} = 0.7$) achieves the best performance across all metrics.

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TABLE IX
IMPACT OF DIFFERENT ATTENTION LAYER CONFIGURATIONS IN SACT MODULE ON PREDCLS TASK

LayerConfig	Recall@K				Mean Recall@K			
Same/Cross	20	50	100	500	20	50	100	500
0 / 1	60.9	76.8	85.7	94.8	43.8	55.9	65.4	78.3
1 / 0	60.8	76.4	85.8	94.2	44.1	56.0	65.5	78.5
1 / 2	61.5	77.5	86.3	95.5	45.4	57.2	66.7	79.4
2 / 0	61.9	77.9	86.8	95.9	46.1	58.2	66.7	80.4
2 / 1	**62.3**	**78.1**	**87.1**	**95.9**	**46.3**	**60.0**	**69.0**	**81.9**

When the target similarity is too low ($\alpha_{\text{target}} = 0.3$), scene-specific prototypes become overly independent of the global prototypes, leading to decreased performance. Compared to this insufficient constraint, increasing the target similarity to 0.7 improves R@20 by 0.3% and mR@20 by 4.2%. This suggests that appropriate knowledge sharing between scenes enhances the model's ability to leverage common relationship patterns. Conversely, when the target similarity is too high ($\alpha_{\text{target}} = 0.9$), scene-specific prototypes become too similar to global ones, causing their decision regions to overlap significantly. This excessive similarity essentially reduces our multiprototype approach to a single-prototype method similar to PE, resulting in performance degradation with R@20 dropping by 0.5% and mR@20 decreasing by 2.4% compared to $\alpha_{\text{target}} = 0.7$.

The optimal value of 0.7 represents a sweet spot that balances scene specificity and knowledge sharing. This indicates that moderate similarity constraints enable scene-specific prototypes to preserve unique scene characteristics while benefiting from the global knowledge, particularly enhancing the recognition of tail categories without compromising the performance on head categories.

5) Transformer Layer Configuration: We investigate the impact of different transformer layer configurations in SACT. To better highlight the effect of contextual interactions and eliminate noise interference from object detectors, we directly classify the relationship features enhanced by SACT in the PredCls task. We compare four configurations by varying the number of intrascene and cross-scene attention layers: 1) 1/0 (only intrascene); 2) 0/1 (only cross-scene); 3) 2/1 (more intrascene); and 4) 1/2 (more cross-scene).

As shown in Table IX, different attention layer configurations exhibit significant performance variations. First, when comparing single-type attention configurations, increasing the number of same-scene attention layers (from 1/0 to 2/0) brings improvements of 1.1% in R@20 and 2.0% in mR@20. This demonstrates that in-depth modeling of scene-internal context is crucial for relationship recognition, especially for enhancing the recognition capability of tail categories.

Second, adding a cross-scene attention layer on top of two same-scene attention layers (from 2/0 to 2/1) further improves performance by 0.4% in R@20 and 0.2% in mR@20. In contrast, the configuration with one same-scene and two cross-scene layers (1/2) performs 0.4% worse in R@20 and 0.7% worse in mR@20 than the two same-scene layer configuration (2/0), suggesting that overemphasizing cross-scene interactions may dilute scene-specific features. The optimal 2/1 configuration outperforms all others across all metrics, showing that relationship recognition benefits from balanced hierarchical processing: first, using multiple same-scene attention layers to fully capture internal context and build strong scene-specific representations; then, enabling moderate cross-scene interaction to share knowledge and gain complementary cues—without losing scene specificity. This design is especially effective at balancing performance between head and tail categories.

TABLE X
COMPARISON OF DIFFERENT PROTOTYPE GENERATION STRATEGIES FOR SSMPNET ON PREDCLS TASK

Method	Recall@K				Mean Recall@K			
	20	50	100	500	20	50	100	500
Direct	61.0	76.6	85.7	95.0	48.9	62.8	71.5	83.4
Hierarchical	**61.5**	**77.7**	**86.8**	**95.3**	**51.6**	**65.4**	**74.0**	**84.3**

TABLE XI
ABLATION STUDY ON IMPACT OF MASKING RATION

ratio ρ	PredCls		SGDet	
	R@50	mR@50	R@50	mR@50
0.1	77.5	63.6	19.2	12.5
0.5	77.6	64.2	19.3	12.8
0.7	**77.7**	**65.4**	**19.5**	**13.0**
1.0	77.7	65.0	19.5	12.8

6) Prototype Generation Method: As described in formulation (28), we propose a hierarchical approach to generate scene-specific prototypes P_s from global semantic prototypes P_{base}. To validate the effectiveness of this method, we compare it with a direct generation approach

$$P_s = \text{MLP}_{\text{direct}}(P_{\text{base}}) = W_{d2}(\text{ReLU}(W_{d1}P_{\text{base}})). \tag{40}$$

This approach applies a two-layer neural network to directly transform global prototypes into scene-specific representations, where W_{d1} and W_{d2} are the learnable parameters for each scene. We evaluated SAMP using this direct generation strategy to assess its impact on performance across different tasks and metrics.

As shown in Table X, the hierarchical generation strategy outperforms the direct generation. The improvements are more substantial in mR metrics (2.7% increase in mR@20 and 2.6% in mR@50), indicating that the hierarchical approach particularly benefits tail category recognition. This demonstrates that explicitly modeling the relationship between global semantic knowledge and scene-specific features through the hierarchical generation mechanism provides a more effective framework for prototype learning than direct transformation.

7) Masking and Filtering Ratios: As described in Section III-D.1, we filter relation features based on their predicted confidence scores to suppress irrelevant relationships. This not only mitigates their interference in subsequent contextual interactions, but also reduces computational overhead.

We conduct experiments using filtering ratios of 0.1, 0.5, 0.7, and 1.0. The results are summarized in Table XI. When the masking ratio is too low (e.g., 0.1), only a small subset of relation features are updated by the SACT module, causing the model to behave similar to using only the SAMP module, thus limiting the benefit of contextual interaction. Conversely, when the ratio is set to 1.0 (no masking), all relations participate in feature interaction, introducing irrelevant noise that degrades performance.

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These comprehensive ablation studies confirm the effectiveness of the scene-aware approach. The unique contextual interaction module reduces noise interference while efficiently transmitting information, and the scene-specific prototypes accurately model the features of different categories, providing a viable approach for solving remote sensing scene graph tasks.

V. Discussion

We observe that remote sensing images often contain multiple scene centers within a single image, such as airports, roads, and urban areas. Based on this, we propose that relationships or events between objects are inherently scene-dependent. Guided by this idea, we design scene-aware interaction modules and prototype matching methods that adapt to different scene contexts. Our approach provides a new way to tackle SGG in remote sensing by building models that respect the unique structure of satellite imagery.

Our method improves performance on medium and tail relationship categories. However, like other prototype-based approaches, this improvement comes with a small drop in performance on head categories. Specifically, mR increases while overall recall (R) slightly decreases. This means the model becomes better at recognizing rare relationships but slightly less accurate on frequent ones. In addition, because we use many transformer layers to model interactions between objects and relationships, the model requires more GPU memory. This may limit its use on devices with limited resources. Future work could explore ways to reduce memory usage or make the model more efficient.

VI. Conclusion

In this article, we introduce the concept of scene attributes of objects and relational pairs in remote sensing images. Based on this concept, we design SSMPNet. The SACT module is developed to facilitate differentiated interaction between objects and relations with varying scene attributes, enabling efficient message passing. Meanwhile, we propose the SAMP module to generate scene-specific subprototypes for relational pairs of different scene attributes and employ diverse optimization methods to ensure a proper distribution of these subprototypes. Our approach significantly alleviates prediction errors caused by the long-tail distribution of datasets and offers an effective method for remote sensing SGG.

However, current SGG methods still have certain limitations. For example, they can only recognize predefined categories specified in the dataset. With the development of remote sensing vision-language models [72], [73] in recent years, the text–image feature matching capabilities of pretrained models have laid a foundation for addressing SGG tasks under open vocabulary settings. Meanwhile, while research on static SGG has made progress, the generation of dynamic remote sensing scene graphs—where relationships between objects change over time—is still in its early stages. Therefore, we encourage more researchers to explore related work from these perspectives.

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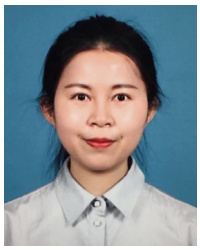
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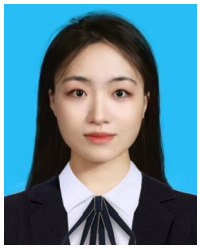
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